

A New Kingdom: Wētā FX Returns to The Planet of The Apes

Philip Leonhardt
Wētā FX, New Zealand
pleo@wetafx.co.nz

Alex Nowotny
Wētā FX, New Zealand
alexno@wetafx.co.nz

Erik Winkvist
Wētā FX, New Zealand
quister@wetafx.co.nz

Paul Story
Wētā FX, New Zealand
story@wetafx.co.nz

ABSTRACT

Featuring dynamic interactions between apes and humans, a world reclaimed by nature, and large-scale combustion and water simulations, Kingdom of the Planet of The Apes is the result of several years of technical advancements. This paper will explore how artists brought the CG apes to life, and the myriads of landscapes that were built, dressed and eventually destroyed.

ACM Reference Format:

Philip Leonhardt, Erik Winkvist, Alex Nowotny, and Paul Story. 2024. A New Kingdom: Wētā FX Returns to The Planet of The Apes. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH Talks '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641233.3665344>

1 DIGITAL APES

Performance capture in the wild was the foundational approach for this film as it provided visually rich environments for the camera and situational context for the actors. A mix of active motion capture and “fauxcap” (using witness cameras) was employed for body motion, with head-mounted cameras (HMC) for recording actors’ facial performances. Witness cameras were crucial in scenarios where actors had to perform outside the range of the active volume or where the HMC boom impeded scene blocking. These were genlocked traditionally operated video cameras, as well as a stereo machine-vision rig mounted to the motion picture camera, providing a front row seat to the action and capable of providing a depth mesh in post for aiding camera tracking and matchmove.

The dramatic increase of spoken dialogue in this film required an efficient facial animation process. The HMC was upgraded to a stereo pair, providing parallax at 48 fps. The additional perspective provided multiple benefits including a derived 3D mesh that validated important depth information, such as the extension of lips or thrust of the mandible, which is difficult to gauge from a single view. This was combined with an advanced version of Wētā’s Facial Deep Learning Solver (FDLS) and used for the first time on an ape character. The solver produced a consistent baseline animation for each shot by taking actor marker data as input and solving keys directly onto a traditional FACS rig. The solver was trained on the

relationship between actor marker data and shape weights on their ape puppet. We used a novel way of training the solver, which required minimal input and could be trained by animators. The brain was able to learn complex non-linear correlations between human and ape muscle movements and even our chosen animation style for each character. Animators could tune the solver with new recipes and shape variations to further advance visual parity with the performer whilst remaining fully controllable for manual fine-tuning.

On top of the FDLS was Wētā’s unique DeepShapes approach, also introduced for the first time on a CG ape character. DeepShapes attempts to mimic the skin’s depth, from fascia to epidermis layers. This approximates temporal details that are not achievable from a traditional blendshape puppet such as memory folds, inertia, secondary overlapping motion, and subtle shape variation depending on a shape’s activation state. They combat the linearity and simplicity of blendshapes with minimal overhead in asset build or animator time, maintaining the benefits of blendshapes but with the visual result of something akin to dynamic scan data.

Biometric data, CT scans, and plaster casts were used as reference to achieve the facial anatomy and topology of each hero ape. Modelers completely reworked the body topology to allow greater definition and anatomically correct movement. By taking advantage of our muscle simulation systems, highly accurate deformations resulted, and per-shot cleanup was significantly reduced.

Every ape was rigged with tissue dynamically solved upon animation publish. Additionally, our GPU-accelerated animation framework, *Koru*, was used for the first time in the *Apes* franchise. The *Koru* supports a far greater number of puppets to be active in a scene while still managing near real-time frame rates. Artists created 105 unique ape grooms, individually sculpting them to capture the look and personality of the characters and their variations (dry, damp, drenched, etc.). A typical hero groom comprised of up to 600,000 strands and 11.5 million vertices. These new physically based fur models were used with our simulation framework, *Loki*. This enabled artists to dial in the properties of ape fur and clothing using real world properties such as Young’s modulus, density, and fur strand diameter, among other attributes. We were also able to simulate a much higher percentage of the grooms, sometimes exceeding 20% of the strands. This enabled complex simulations such as wet hair, which requires clumping and cohesion of many strands instead of sparse control curves. Using a physics-based solver enabled realistic fur interactions with the elements, from small flutters to strong winds as well as water interaction. Using a custom wind graph gave full control of gusting and timing with

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Talks '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0515-1/24/07

<https://doi.org/10.1145/3641233.3665344>

added 3D fluid turbulence. The new *Loki* solver, as well as advancements in the tissue simulation pipeline and tooling, increased workflow efficiency dramatically while delivering a more realistic result.

To further optimize workflows, we relied on our latest in-house shading and rendering architecture, *Squid*. Previously, pre-compiled shading nodes were passed to the renderer as part of a network. By contrast, *Squid* is reliant on just-in-time code-generated and compiled shaders, providing artists the ability to build complex networks that can be in-lined and optimized at publish time. *Squid* provides a Universal Scene Description (USD)-based referencing architecture, allowing artists to template everything that's shared across multiple asset types, and work on individual cases as a delta state. This allowed us to quickly prototype and deploy updates, which encompassed shader parameters, attributes, primitive variables, textures, looks, etc. The approach resulted in a considerable reduction in shading storage and overall render times, which in turn gave shading, texture and lighting artists more time to iterate.

2 ENVIRONMENT & AMBIENT MOTION

It was important in this post-apocalyptic world to depict nature having fully taken over. Wētā's in-house dressing tool, *Scenic Designer*, allowed for large landscapes such as the skyscraper ruins, to be dressed in foliage procedurally, using over 100 plant species. Each of the ruins were modeled and dressed for a unique appearance, so simple rotations of instanced assets provided a great deal of look variety with relatively few exemplars. A more detailed dressing pass took into consideration natural factors such as the amount of available light, moisture, wind, and terrain topology to inform dressing choices. To breathe life into these digital sets, artists simulated looping ambient wind motion with *Loki* based on Beaufort scale strengths for each plant.

We used *Houdini*'s Procedural Dependency Graph to manage the motion data, which was then assigned by layout artists. Plant interaction FX were critical for the storytelling across the film, particularly during a cat-and-mouse like sequence, set in a field of tall grass. The reeds were modeled using our procedural plant toolset, *Lumberjack*, which provided an internal curve structure that served as a skeleton to deform the model. Per-CV attributes for stiffness, density, and width described the material properties of each branch or leaf. FX artists simulated ape interactions against small sections of grass which were designed to seamlessly integrate with the larger layout, resolving collisions with high precision. Finally, pre-cached ambient motion simulations were applied with modulated amplitudes and time offsets to give the impression of wind waves traveling across the field.

3 EFFECTS SIMULATIONS

This story required water and combustion FX simulations of varying complexity. One such challenge was a river sequence where a body of violent CG water had to seamlessly blend into plate water photography. Studying reference footage of rapids informed the riverbed topology required to create a consistent flow of water with swells and specific features. We created multi-resolution simulations which could be blended depending on the camera angle. To maintain art directability, we further developed our *Loki* render

graph to mix caches and blend Fast Fourier Transform (FFT)-based surface flow approximations with existing simulations. Individually generated water features were integrated into the overall river simulation. A simplified geometry representation of this surface was then shared with animators as a proxy. Once the animation was approved, a hero simulation was triggered to account for the ape's long hair against the flow. This result then fed high detail simulations of water thin film and spray elements over the fur.

The third act silo flood sequence began with a dramatic explosion simulated with layers of sand, gravel, and molten metal debris, which were all combined to depict a pressurized termite reaction. To achieve the complex choreography between the flooding silo and CG apes, we used a similar approach to the river, with proxy water used by animators during their blocking stage to maintain physical plausibility. We then ran high-resolution simulations under similar boundary conditions to achieve levels of detail that allow for realistic interaction with the characters and environment. Developers used this opportunity to refine and improve the collision handling of various states of simulations in these specific cases. Some shots required extremely high-resolution fluid simulations to support cameras at water level. Foam simulations and representations were also revised to account for grime and silt that accumulated as flood waters pushed deeper into the silo.

FX artists created combustion simulations of varying complexity, ranging from small torches to a flaming village, complete with collapsing burning structures. Once more, *Loki* handled simulation duties using chemical formulas and real-world compounds to replicate the array of fuel sources. *Loki*'s ability to approximate accurate thermodynamics enabled artists to match the look and behavior of on-set fires, allowing them to address artistic rather than technical notes. This framework produced predictable real-world values which were rendered with our spectral renderer, *Manuka*, which in-turn, also incorporated their emission spectra to illuminate other elements in the scene.

4 LENS WORK

A mix of 1970s Panavision C-series and modern T-series primes contributed unique optical artifacts which needed to be matched in rendered elements. We profiled the 26 lenses used to measure their optical characteristics. For the first time, we measured lens transmission to determine attenuation across the visible spectrum and accounted for this in renders. Lens distortion and anamorphic compression were measured as a function of depth through the novel use of a mocap calibration wand for extremely accurate models for each lens. A Nuke lens package applied these characteristics to rendered elements for integration into photography and a highly convincing visual result.

ACKNOWLEDGMENTS

Matt Fitzgerald, Emiliano Padovani, Angelo Perrotta, Jordan Cario, Frederic Valleur, Caitlin Sue, Claude Schitter, Kevin Kelm, Stuart Adcock, Peter Hillman, Allison Orr, Dejan Momcilovic, Alexey Stomakhin, Arthur Graff, Wētā FX, Wes Ball and 20th Century Studios.